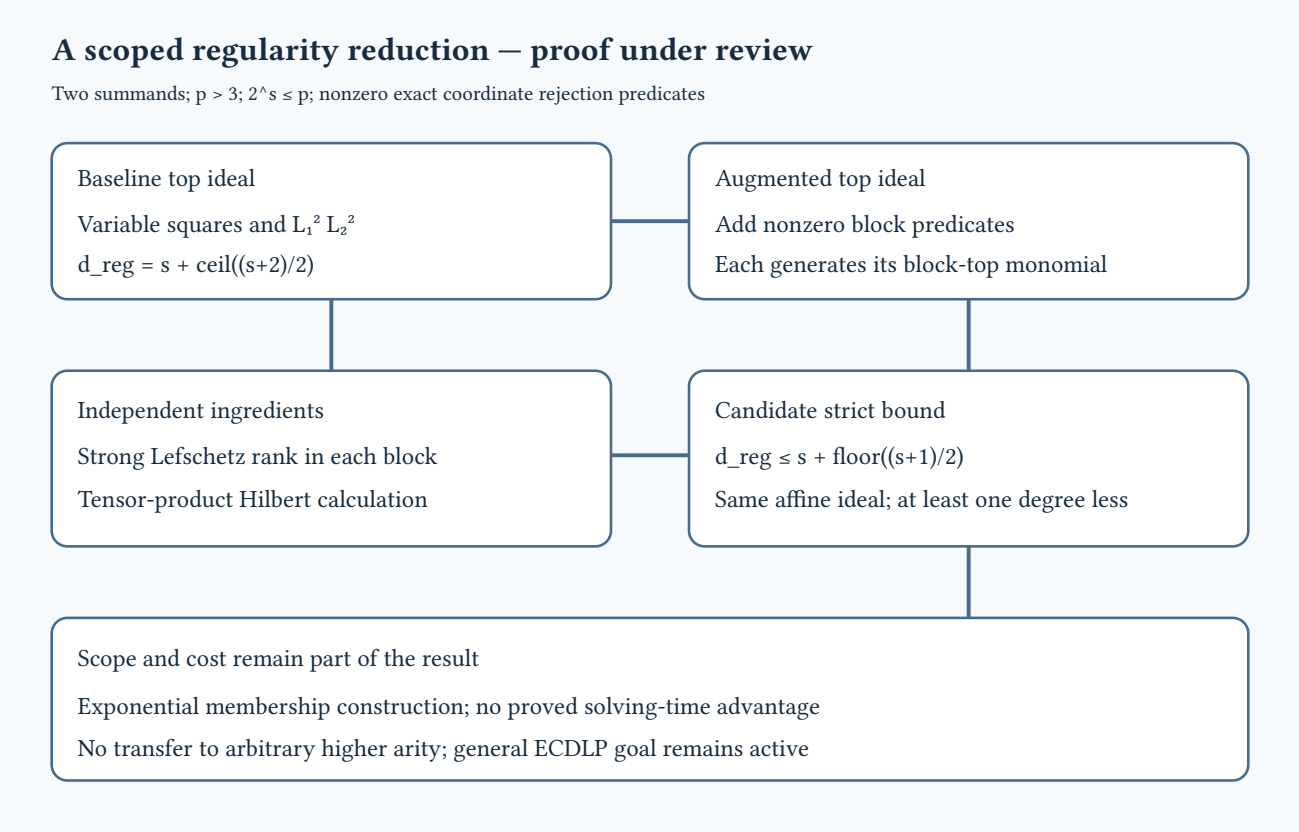
A strict regularity reduction for two-summand prime-field systems

Derived proof; independent adversarial review pending

This report accompanies `proof.md` and `proof-status.json`. It states an all-parameter algebraic claim with explicit assumptions. No scientific trials were run for this derivation. The broader goal of a general ECDLP improvement remains active.



Exact claim

For a nonsingular curve over a prime field of characteristic greater than three, fix a finite rational target with nonzero ordinate. Encode each of two candidate abscissas by an injective affine map from s Boolean bits, with nonzero weights and two to the power s at most p.

Let F0 contain the Boolean equations and the S3 quartic. Let F1 also contain the exact multilinear rejection predicate for each coordinate domain, and assume both predicates are nonzero. Then

$$d_{\mathrm{reg}}(F_0) = s + \left\lceil \frac{s+2}{2} \right\rceil$$

$$d_{\mathrm{reg}}(F_1) \le s + \left\lfloor \frac{s+1}{2} \right\rfloor = d_{\mathrm{reg}}(F_0) - 1$$

The two lists generate the same affine ideal. The invariant is the degree where the ideal generated by the **highest homogeneous parts of the supplied generators** fills the homogeneous component. It is not an invariant of the affine ideal alone.

Why the calculation works

Modulo the variable squares, the quartic's top part is the product of the squares of two weighted block-linear forms. Strong Lefschetz gives every block multiplication rank. Tensoring those maps gives the exact Hilbert series and the baseline threshold.

Any nonzero homogeneous squarefree form in one block generates its full top monomial: multiply by the complementary variables of a monomial having nonzero coefficient. Every other term dies modulo the variable squares. Adding a nonzero predicate in each block therefore removes enough high multidegrees to lower the vanishing threshold by at least one.

Rationality is proved separately. An explicit line through the negative target realizes both abscissas as rational curve points whenever S3 vanishes. Hence the membership predicates already vanish on all Boolean solutions. The Boolean coordinate ring is a product of copies of the base field, so equality of these zero sets also gives equality of the generated affine ideals.

General algebra lemma

With m disjoint blocks of s variables, and a generator which is the product of the e-th powers of their weighted linear forms, assume every weight is nonzero, one at most e at most s, and characteristic zero or greater than s. The exact baseline is

$$(m-1)s + \lceil \tfrac{s+e}{2} \rceil.$$

Adding one nonzero block predicate in every block gives an upper bound smaller by at least m minus one. This is an algebra lemma. It does not prove that higher-arity Semaev systems can receive rational-membership predicates without changing their solutions.

Construction and limitations

A fixed domain with identically zero rejection predicate does not satisfy the strict-reduction premise. A constant-one predicate gives trivial inconsistency. For a newly selected window on an odd-prime-order curve, a rejected starting coordinate exists by Hasse's bound; this does not authorize changing a pre-existing fixed problem.

Each distinct membership table needs two to the power s character tests, order s times two to the power s interpolation additions, and order two to the power s stored field elements. Ordinary exponentiation gives order two to the power s times log p field multiplications for the character tests. These costs remain charged.

The bound still grows linearly with s. No solving-degree reduction, last-fall-degree reduction, solver speedup, higher-arity elliptic transfer, unknown-scalar recovery, novelty, or formal proof-assistant verification follows from this note. Earlier finite experiments remain under their own validation status.

Retrieved sources and review

- Caminata and Gorla, Definition 1.4: arXiv:2112.05579v2.
- Phuong and Tran, Theorem 2.7: arXiv:2211.13548.
- Kosters and Yeo, summation-polynomial definition: arXiv:1503.08001.
- Sutherland, Hasse's theorem, Theorem 7.3: MIT 18.783 Lecture 7.

The prospective review plan requires a fresh independent derivation to be recorded and hashed before the reviewer sees the immutable producer proof. Review status is pending in this producer snapshot; any later verdict is a separate artifact.